STATEMENT OF WORK
02 APRIL 2025

Project #: LXEZ201076
Project Title: Repair Water Lines, Phase II
Facility: Kadena AB
Installation: Kadena Air Base, Okinawa, Japan
Preparer [redacted]
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Table of Contents

Contents

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1 GENERAL INFORMATION

1.1 LEVEL OF CONSTRUCTION

This project is for the design and construction of facilities defined as Permanent Construction per UFC 1-200-01, paragraph 1-3.2.

1.2 PROJECT SUMMARY

The project scope consists of, but is not limited to, providing all field investigations, surveys, design, plant, labor, supervision, materials, equipment, supplies, transportation, third-party services, and incidental related work to design and execute construction work defined in this Statement of Work and the attached documents. The work includes all necessary finishing work to provide the Government with a final product that is fully functional and leaves no areas of the facility that appear incomplete or unfinished.

The proposed work in Repair Water Lines Around the Airfield, Phase II primarily focuses on the rerouting and replacement of the existing water mains that are located underneath the airfield runways. Replacing the old deteriorating water mains requires the installation of new fire hydrants along the proposed alignments in accordance with the Unified Facilities Criteria (UFC) and National Fire Protection Association (NFPA) requirements as well as providing water main connections onto existing distribution mains and service lines that feed the current facilities.

The existing water lines are at the end of their life cycle, being over 60 years old, and are prone to high rates of leakage and unplanned ruptures. The original water main network runs through the middle and at both ends of the runways, which puts the airfield operation at risk of closure if those segments require repairs.

1.3 PROJECT REQUIREMENT

This is design-build project with the purpose of replacing the existing water distribution mains. This project will include replacing the 16-inch and 12-inch water mains along a new route within the roadway right of way to eliminate the distribution mains running through the flightlines. New fire hydrants will be installed along the proposed alignments in accordance with the UFC and NFPA requirements, providing water main connections onto existing distribution mains and service lines that feed the current facilities, and providing new pressure reducing valve systems to balance the water distribution network's pressure and flows. Proposed repairs to lines running through the runways include trenching the existing mains (abandon in place) and filling with flowable structural concrete, then backfill to existing grade with excavated material. Other existing mains will be abandoned in place. This project also includes SCADA to enable continual monitoring ensuring water quality is constant and meets requirements. A Cultural Assessment Survey will be conducted on all new water main routes. Work for the project is described in more detail in Section 8 of this Statement of Work (SOW). Sustainable principles, to include life cycle cost- effective practices shall be integrated into the design, development, and construction of the project(s) in accordance with the latest version of UFC 1-200-02, High Performance and Sustainable Building Requirements.

Construction activities may commence after the approval of the Environmental Protection Plan (EPP) and AF Form 103 (work clearance). The Contractor shall coordinate on-site safety requirements through the 718th Civil Engineer Squadron (718 CES).

1.4 GOVERNMENT FURNISHED EQUIPMENT/MATERIAL (GFE/GFM)

The following GFE/GFM will be provided:

a. No GFE/GFM will be provided.

2 SITE CONDITIONS

2.1 WORK SITE LOCATION

. The proposed work shown on Figure 5 and described below is conceptual. Pipe sizing, alignment, location of PRVs, and all new work shall be determined during design based of field conditions, current UFC requirements, and all other applicable criteria. The new work shall also incorporate water system improvements to be constructed under Project AF406, Phase I, based on project site conditions and design information.

- 12" C900 DR14 water piping routes include piping from Omaha Avenue & Stethem Street PRV stations, following Walker Road around the North Flight Line, to Collision Drive PRV Station.
- From Collision Drive PRV Station, the 16" DR14 C905 Water Main to Molland Street.
- Also includes the 16" C905 DR14 water main following Schreiber Avenue, to Douglas Boulevard, to Kuter Boulevard, down to Molland Street.

a. Affected facilities / infrastructure are:

(1) Full list of facilities impacted will need to be determined by the contractor during design and construction. This will include any water outages, power outages, road closures, or flightline closures necessary during design survey work or during construction.

(2) The contractor must coordinate with 18 CES Water Shop if water lines will be interrupted.

(3) The contractor must coordinate any flight line activities through the proper TACW process.

(4) Any road closures must be coordinated to 18 Security Forces through 718 CE.

b. Occupancy of Premises

(1) Government personnel may occupy surrounding facilities not under active construction.

(2) The Contractor will have full access during authorized working hours and shall be responsible for securing the construction area daily.

c. Lay Down Area: Contractor will be responsible to coordinate and obtain approval of lay-down area(s) to store materials and equipment and oversee the construction of the proposed infrastructure improvements.

2.2 AS-BUILT DRAWINGS

See attached drawings as a reference.

The Government bears no responsibility for the accuracy or completeness of these drawings. Contractors shall conduct field investigations of the affected facility to ascertain the actual existing conditions prior to beginning the design. The Contractor is solely responsible for any errors, omissions, delays, or additional costs attributed to the failure to do so.

2.3 REGULATED BUILDING MATERIALS

. Asbestos Containing Material / Lead Base Paint (ACM/LBP)

a. The project may require sampling and testing of ACM/LBP. The Japan Environmental Governing Standards (JEGS) will be followed during the sampling process and testing. See Attachment-9a Section 01 57 19.01, Temporary Environmental Controls – Kadena Air Base

2.4 USE OF LATRINE AND UTILITIES

The Contractor shall be allowed to use the existing latrines of facilities listed during standard and non-standard workdays and hours.

Contractor shall be required to conduct final cleanup of any areas used or occupied

The Contractor shall be allowed to use the existing latrines of facilities listed during standard and non-standard workdays and hours.

Contractor personnel during the performance of the project. Location of the latrine(s) will be designated and approved by the Contracting Officer (CO) and Contracting Officer Representative (COR).

Contractor furnished latrines shall be removed from Government property within three calendar days after final acceptance of the project. Contractor may use all approved and available services (i.e., electricity, water, and sewage) at the construction site.

The Contractor shall request COR approval prior to making connection to existing facility utilities. any areas used or occupied by Contractor personnel during the performance of the project. When latrines are not available in the work area, the Contractor shall be responsible for providing latrines for his personnel.

3 PROJECT PHASING & SCHEDULE

3.1 PERIOD OF PERFORMANCE

. The Period of Performance (POP) allowed on this contract is detailed below. The Contractor shall be required to complete the entire work, ready for use. This includes pre-construction and material submittals, government reviews, material procurement, unfavorable weather, and construction time. The time for completion shall include final cleanup of the premises.

Days for Design:	271	calendar days
Days for Construction:	540	calendar days
Days for Administration:	30	calendar days
Total POP	841	**calendar days**

a. Notice to Proceed. Day one of the POP is the day after the day the Contracting Officer issues the Notice to Proceed (NTP) on the design. If no serious issues are identified during the Pre-Design Conference, the Contracting Officer normally issues the NTP at the end of the conference.

b. Close-Out. All requirements of this contract including the design, submission and Government review of submittals, resolution of design issues, material acquisition, on-site construction work, site clean- up and return of the facility, ready for use, must be completed by the last day of the POP. Refer to section 01 78 00 for suspense of Closeout Submittals.

3.2 PRE-DESIGN MEETING

. Within 10 days of receiving the notice of award, the Contractor will schedule a Pre-Design Meeting with the Contracting Officer, Civil Engineer Squadron (CE) Project Delivery Team, and other project stakeholders.

a. At the Pre-Design Meeting, the Contractor shall submit a Project Schedule for review by the Government and re-submit updates, as necessary, until project completion.

b. During the meeting, the Contractor may ask any questions pertaining to the contract, including questions pertaining to the design, progress schedule, or other issues. If requesting changes to any elements of the contract, submit a written explanation justifying the change(s), accompanied by diagrams, technical data, and any other justifying documents to the Contracting Officer. The Government recognizes only those changes agreed upon in writing by the Contracting Officer as official changes to the contract.

3.3 DESIGN

. Design Schedule

(1) The design work shall be completed in accordance with the following sequence/timeframe. Written approval from CO is required to proceed to the next level of design submittal. Contractor shall attend review meetings as scheduled.

DESIGN SCHEDULE	SHALL BE COMPLETED WITHIN	
Submit 35% Design Documents + Survey	90	calendar days after NTP
Government 35% Design Review	14	calendar days after submission
35% Design OBR	7	Calendar days after design review
Submit 65% Design Documents	30	calendar days after 35% design OBR
Government 65% Design Review	14	calendar days after submission
65% Design OBR	7	Calendar days after design review
Submit 95% Design Documents	30	calendar days after 65% design OBR
Government 95% Design Review	14	calendar days after submission
95% Design OBR	7	Calendar days after design review
Submit Revised 95% Design Documents	30	calendar days after 95% design OBR
Government Revised 95% Design Review	14	calendar days after submission
Revised 95% Design OBR	7	Calendar days after design review
Submit Final Revised 95% Design Documents	7	calendar days after Revised 95% OBR
Design Subtotal:	271	**calendar days [Make sure that the design duration is consistent with 'Days for Design' in section 3.1]**

a. Design Review Meetings

(1) The Contractor shall attend meetings with the Government at the 35%, 65%, 95%, and Revised 95% design stage to discuss details of the design with Government stakeholders. Submit all required design documents just prior to, or at the onset of each meeting.

(2) The Contractor shall prepare all meeting minutes to document both party's comments and distribute copies to attendees via email within 3-working days after each meeting.

(3) The Contractor shall provide a proper resolution for all legitimate issues brought up by Government reviewers and return the completed PACAF Form 225 within 14 days.

3.4 CONSTRUCTION

. Pre-Construction Conference: The Contractor shall request a pre-construction conference with the Government no later than 35-calendar days prior to the start of on-site work.

a. Construction Schedule:

SITE WORK/CONSTRUCTION SCHEDULE	SHALL BE COMPLETED WITHIN

a)	Material Submittals:	60	calendar days after Final Revised 95% design approval
b)	Government Material Review:	30	calendar days
c)	Procurement of Materials:	120	calendar days
d)	Construction Time:	315	calendar days
e)	Unfavorable Weather:	15	calendar days
Construction Subtotal:		540	**calendar days [Make sure that the construction duration is consistent with the "Days for Construction" in section 3.1]**

b. Follow the working hours and days listed in the specifications. To wit, "Normal Construction Working Hours and Work Days shall be from 07:30 to 18:00, Monday through Friday, excluding U.S. Federal holidays."

(1) Work outside of normal hours or days is prohibited without written approval of the Contracting Officer.

(2) Federal Holidays are listed in the specifications.

c. Weather Delays: The Contractor shall provide a weather delay request within 10 days of unfavorable weather that caused the work delay per FAR 52.249-10(b)(2).

d. Start Work Notification

(1) No less than 7 days prior to the start of on-site work, the Contractor shall inform the Contracting Officer and CE PM in writing of their intended start date. The Contractor must have a fully coordinated the Work Clearance Request (AF Form 103) and have received approval of the Environmental Protection Plan (EPP) prior to being authorized to start on-site work.

(2) See specifications for details concerning the AF Form 103.

(3) See Section 01 57 19, TEMPORARY ENVIRONMENTAL CONTROLS, for details about drafting and submitting the EPP.

4 DESIGN REQUIREMENTS

4.1 DESIGN CRITERIA

. See specification section 01 33 16.00 10, DESIGN DATA AFTER AWARD.

a. The Designer of Record (DOR) bears full responsibility to incorporate all applicable criteria into the design of this project, whether specifically referenced in this text or not.

b. Government review and approval of Contractor Submittals and Contractor Design Drawings does not relieve the Contractor of the responsibility to provide technically accurate and complete designs or materials. The Government shall not be responsible for any additional costs that could have been avoided through an appropriate contractor design.

5 SUBMITTAL REQUIREMENTS

5.1 GENERAL

. See specification section 01 33 00, Submittal Procedures; and section 01 78 00, Closeout Submittals.

a. Submittals for all materials and shop drawings must be coordinated and approved by the 718th Civil Engineer Squadron (718 CES) Project Manager.

b. Project design will be coordinated with the 18th Contracting Squadron (18 CONS), the Using Agency, 718 CES and other agencies involved in project reviews.

c. The Contractor's licensed Architects and Engineers shall stamp, date, and certify all professional design documents. Provide proper credentials prior to beginning design work. Construction documents must be stamped, signed and dated prior to construction and upon completion of the project.

d. For civil work, perform and provide calculations for the proposed water system in accordance with UFC 3-230-01, UFC 3-230-03, and UFC 3-600-01. Design calculations should include modeling the entire water system to ensure adequate flow and pressure area maintained and impacts to water quality are reduced or eliminated. While the project will only repair a portion of the water system, the entire water system will be impacted by the proposed work.

5.2 SUBMITTAL DOCUMENT FORMAT

See specification section 01 33 16.00 10, Design Data (DESIGN AFTER AWARD); section 01 78 00, Closeout Submittals; section 01 78 23, Operations and Maintenance Data; and section 01 78 24.00 10, Facility Data Requirements.

a. Submit all design documents at each stage of the design and acceptance process.

b. Provide two sets for each submittal: one set for 18 CONS/PKC and one set directly to the 718 CES/CENM flight managing the project.

c. A set consists of one printed hard copy of each submitted document and a CD-R, DVD-R, or CD- ROM containing all of the documents.

d. Do not use media recorders that burn discs regionalized for Japan only. Media must be regionalized for the United States or not regionalized at all.

e. Hard copies shall be submitted per the following schedule:

	Submittal	35% sets	65% sets	95% sets	Revised 95% sets
1	B1 Drawings (full size)	1	1	1	1
2	Specification Package with AF IMT 66	2	2	2	2
3	Annotated Design Review Comments	1	1	1	1
4	Design Analysis/Calculations	2	2	2	2
5	CD-R (all contract documents)	2	2	2	2
6	Draft DD Form 1354			2	
7	Draft Progress Schedule	2	2	2	2
8	Approved EPP			1	1
9	Detailed Cost Estimate (Contractor's bid)	1	1	1	
10	ACM/LBP Report (As Required)	1	1	1	1

6 OTHER REQUIREMENTS

6.1 GENERAL

The project engineer, the user, and 18 CONS/PKC representatives shall accompany the Contractor to conduct joint job site inspection prior to initiation of the site work.

7 PRE-FINAL INSPECTION

The Contractor shall conduct a pre-final walk-through inspection in the presence of the COR and CE Project Manager and publish the pre-final inspection findings in a pre-final inspection report. The pre-final inspection shall be conducted a minimum of 10 calendar days before the final inspection date. A Beneficial Occupancy Determination (BOD) may be required for certain areas if it is in the best interest of the

Government.

7.1 FINAL INSPECTION

. The final inspection report shall certify that all Contractor construction contract requirements have been met.

a. The Contractor shall complete a DD Form 1354, Transfer and Acceptance of DoD Real Property, for Government approval. Submit a draft version of the DD Form 1354 with the 95% design documents. Submit an interim DD Form 1354 at the pre-final inspection and a final DD Form 1354 prior to final invoice. Prepare the DD Form 1354 in accordance with UFC 1-300-08, Criteria for Transfer and Acceptance of DoD Real Property.

8 ENVIRONMENTAL COMPLIANCE CRITERIA

8.1 GENERAL ENVIRONMENTAL COMPLIANCE

. The Contractor shall draft and submit an Environmental Protection Plan (EPP) per specification section 01 57 19.01 30, TEMPORARY ENVIRONMENTAL CONTROLS, and comply with the plan during the entire period of performance.

a. Be advised that on-site construction work cannot begin until Government approval of the Contractor's EPP. The AF Form 103 Work Clearance Request will not be approved until the EPP is approved.

(1) Submit the above items early to avoid a delay in the construction schedule. The Contractor is solely responsible for any delays attributed to the Government's disapproval of plans that do not meet the contract requirements.

b. REQUIREMENTS FOR HAZARDOUS BUILDING MATERIALS

c. For unexpected discovery of hazardous building materials, comply with specification section 01 57 19.01 30.

9 DETAILED DESCRIPTION OF WORK

9.1 Repair Water Lines, Phase II LXEZ201076CR

9.1.1 Civil:

(1) Temporary erosion control and installation of fences for red soil, especially airfield grass area.

(2) Replace the water mains along a new route within the roadway right of way, to eliminate the distribution mains running through the flightlines.

(3) Install new water mains as generally shown on the conceptual layout. Final pipe size and layout to be confirmed during design.

(4) New connections to pump station(s) and reservoir. Please note: the connection points indicated on Figure 5 are conceptual. Obtain final approvals from 718 CES and Contracting Officer during design for all required water system connections.

(5) Proposed improvements shall result in water system pressure that meet UFC 3-230-01 requirements.

(6) Provide new water main connections onto existing distribution mains and service lines that feed the current facilities. Please note: the connection points indicated on Figure 5 are conceptual. Obtain final approvals from 718 CES and Contracting Officer during design for all required water system connections.

(7) Provide new Pressure Reducing Valve system(s) and all associated appurtenances to balance the water distribution network's pressure and flow.

(8) Proposed work to existing lines running through the runways includes:

i) Trenching to existing mains (abandon in place).

ii) Fill existing pipes to be abandoned with CLSM.

iii) Backfill to existing grade with excavated material.

iv) Include removal of hydrants, valve boxes, and any other surface features for existing utilities to be AIP. If applicable, include removal, crushing of vault floors, backfill, etc. for any large concrete vaults.

(9) Other existing mains are to be abandoned in place after the new system is operational.

(10) SCADA (Supervisory Control and Data Acquisition) to be installed to allow continual monitoring of the water system and ensure water quality is constant and meets requirements.

(11) Geotechnical Report will be required: One 10-foot bore every 500-feet along the water main alignment.

(12) A subsurface geotechnical investigation is recommended along the proposed water main alignment, along with detailed topographic survey which includes utility mapping and potholing at all proposed water main/existing utility crossings and tie-in points.

(13) Contractor shall restore all affected/damaged structures including paved/unpaved ground surfaces to its original or better condition with no additional cost to the Government.

(14) Provide all fittings, accessories, and appurtenances required for a complete and usable system. Valves shall be provided in accordance with UFC 3-230-01 and UFC 3-600-01.

9.2.1 Environmental

(1) Cultural Assessment Survey will be conducted on all new water main routes.

(2) UXO Survey and mitigation shall be conducted on all new water main routes.

(3) Sustainable principles, to include life cycle cost-effective practices, will be integrated into the design.

(4) Flood Plain Determination: The waterlines associated with this project fall within the 100yr floodplain. The Floodplain Management Rule applies.

(5) Stormwater Pollution Prevention Permit is required.

9.3.1 Fire Protection Requirements

(1) Upgrade the water system to provide fire flows and fire water demands IAW UFC 3-600-01 and facility fire suppression system requirements.

(2) Existing water distribution network lacks adequate firefighting flows and pressures along the north flightline: one of the main objectives of this project is to ensure the new water mains, PRVs, and associated appurtenances meet the flow and pressure requirements and demands.

(3) New fire hydrants will be installed along the proposed alignments in accordance with the UFC and NFPA requirements.

(4) This project requires the design, review, and oversight services of a Qualified Fire Protection Engineer (QFPE) in accordance with UFC 3-600-01 Fire Protection Engineering for Facilities Paragraph 1-7.

10 ACM / LBP-RELATED WORK

10.1 General

. The Contractor shall ensure all ACM/LBP related assessments, surveys, sampling, reporting, abatement plan design and execution, and waste disposal is conducted by trained and certified in accordance with the applicable Code of Federal Regulations (CFR) and the requirements in the specifications and supplements identified in the

SOW.

10.2 Abatement of Identified Materials

a. General contractor shall review Kadena AB existing water system drawings to be aware of any asbestos-cement-pipe (ACP) in the project work area. Contractor shall properly manage, handle, remove, and dispose of the ACP according to base requirements.

b. The general contractor is required to use a certified asbestos inspector to test for asbestos and/or lead-based paint (LBP) for suspect materials and/or paint which will or may be disturbed in existing infrastructure. Proper planning procedures will be used during the demolition to ensure safety of workers and environment. Proper disposal of any demolition/removal of debris is required. If applicable, the contractor will submit a sampling plan for review to SGXB and CEIE.

c. During design and construction, a Safety Data Sheet (SDS) or new paint products will be submitted scheduled is to be used during construction. The contractor will ensure low volatile paint and primer that does not contain isocyanates and/or lead is used.

d. The general contractor is also required to obtain prior coordination and approval from SGXB on the use of any equipment containing radioactive material (RAM). Examples include LBP XRF analyzer, soil compaction/density tester.

11 ATTACHMENTS

1. Specifications for General Requirements and Existing Conditions
2. Technical Specifications
   a. Kadena Air Base CAD Standard
   b. KABI32-2001 – Fire Prevention and Protection
   c. JEGS
3. Additional Environmental Specifications
   a. 18WG 01 57 19.00 30 Temporary Environmental Controls KAB Supplement (2017-06)
   b. 01 57 19.01 30 Temp Env Controls (2024-12)
   c. UFGS 0157 19.01 EPP Spec (2020-03-03)v2.1
   d. 18 WG 02 83 00, Lead Remediation (2019-08)
   e. UFGS 02 82 00 ACM (2019-08)
   f. 02 82 00.00 30, ACM SUPPLEMENT (2024-12)
   g. UFGS 02 83 00 LBP (2019-08)
   h. 02 83 00.00 30, LEAD SUPPLEMENT (2024-12)
   i. 02 84 16.01 30, Lamps-Ballasts (2024-12) v1.7
4. Kadena Design Guide [or Installation Facility Standard]
   a. Kadena AB Design Guide
   b. KAB Design Guide Appendices V1
   c. KAB Design Guide Appendices V2 Pt. 1
   d. KAB Design Guide Appendices V2 Pt. 2
5. Work Site Location Maps
6. As-Built Drawings
7. Special Contract Requirements
8. Other Contract Requirements

9. ACM/LBP Test Results
10. Quality Assurance Surveillance Plan (QASP)